TOML 2026-27
Maximum Likelihood Estimation

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October 2026

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Chapter 1

Maximum likelihood estimation

The following three chapters discuss basic concepts on estimation theory and basic machine learning, both from a frequentist and a Bayesian perspective. These topics are covered with much more detail in the following: G. Casella, R. Berger, "Statistical Inference" (a classical book on frequentist inference), G. James, D. Witten, T. Hastie, R. Tibshirani, "An introduction to statistical learning" (an introduction to machine learning with examples of code in Python), D. Mackay, "Information Theory, Inference, and Learning Algorithms", (wonderful book on Bayesian estimation, information theory, coding, etc, available online for free in `https://www.inference.org.uk/itprnn/book.pdf`), "Elements of Statistical Learning" by T. Hastie, R. Tibshirani, J. Friedman (a reference book on machine learning), "Pattern Recognition and Machine Learning" by C. Bishop (Bayesian machine learning), and "Deep Learning" by I. Goodfellow, Y. Bengio, and A. Courville (a reference on deep learning, today maybe a bit outdated given the fast advances in the field, online free version in `https://www.deeplearningbook.org/`). We also use some material from "Hands-on Bayesian Neural Networks - A Tutorial for Deep Learning Users", L. V. Jospin et.al. IEEE Computational Intelligence Magazine, May 2022. You can check many of the ideas discussed on neural networks using `http://playground.tensorflow.org/`.

1.1 Modeling, estimation, and prediction

In the first part of the course, we studied probabilistic *modeling problems*, in which the parameters defining a given probability distribution were assumed to

be known, and our problem was to estimate the probability of a given event (i.e. a given output). For instance, assume that we have a coin with a *known* probability of head $p(1) = \theta$, (we represent "head" as "1" and "tail" as "0"). Our goal was to find an expression for the probability that, in an experiment consisting of m independent coin tossing, the observed outcome was $(1, 0, 0, 1, 0, 0, 0, 1, 0, 0)$, i.e. $p(1, 0, 0, 1, 0, 0, 0, 1, 0, 0) = \theta^3(1-\theta)^7$.

The *estimation problem* is the reverse: We know the result of several independent samples generated by a given probability distribution with unknown parameters, and our goal is to give an estimate of these parameters. Following our example, we have a coin with an unknown head probability given by the parameter θ. We perform $m = 10$ independent experiments consisting of tossing the coin and obtaining the result $(1, 0, 0, 1, 0, 0, 0, 1, 0, 0)$. Our problem is to estimate the value of θ. Here we have two possible approaches. An approach would be to give a single value for our estimated parameter, for example, $\hat{\theta} = 0.3$. Another possible answer would be to give a probability distribution that quantifies our degree of belief in the different possible values of θ, for example: $p(\theta|(1, 0, 0, 1, 0, 0, 0, 1, 0, 0)) = B(\theta; 4, 8)$, where $B()$ is the Beta distribution.

The *estimation problem* is very much related to *generative models*, such as, for instance, the generative AI models used to generate text or images that reassemble written text or natural images.

The *prediction problem* is closely related to the estimation problem. We know the result of several independent samples generated by a given probability distribution with unknown parameters. Our goal is to give a prediction of new unseen outcomes generated by the probability distribution:

$$p[y_{11}{=}1|(y_1, ..., y_{10}){=}(1, 0, 0, 1, 0, 0, 0, 1, 0, 0)]. \tag{1.1.0.1}$$

This is very closely related to the goal of *supervised machine learning*. More specifically, we will define supervised machine learning in terms of joint random variables (X, Y), and the goal is to predict a new outcome y, given that we know the corresponding x and a data set of previous pairs (x, y). To make this prediction, we can use either the single value for our estimated parameter, or take into account all possible values of this estimated parameter weighted by their probability (i.e. apply total probability).

In most practical problems, there are several possible models that can be used for estimation or prediction tasks. *Model selection* is the problem of deciding which is the most adequate.

1.2 Two approaches to estimation: frequentist and Bayesian estimation

As we have mentioned previously, there are two main approaches to solving the estimation problem we just described. One uses what is called *frequentistesti-*

mation , which estimates the parameter $\boldsymbol{\theta}$ as a function of the observed data. The second method is to use *Bayesian estimation*, which estimates the degree of belief in the values of the parameters given the results of the experiments.

1.3 Frequentist estimation

We assume that a random variable Y has a PMF (probability mass function) or PDF (probability density function) given by $p_{\boldsymbol{\theta}}(y)$, which is parametrized by $\boldsymbol{\theta}$, the parameter we want to estimate. We sample Y in m independent experiments, obtaining a vector of observed data $\boldsymbol{y} = (y_1, ..., y_m)$.

Frequentist point estimation estimates the value of the parameter $\boldsymbol{\theta}$ by means of a function T of the observed data $\boldsymbol{y}$. *This function must depend only on the observed data and must not contain in its definition the values of the parameter* $\boldsymbol{\theta}$. We say that T is an *estimator* of $\boldsymbol{\theta}$, while $T(\boldsymbol{y}) = \hat{\boldsymbol{\theta}}$ is an *estimate* of $\boldsymbol{\theta}$. However, very often we will also use the symbol $\hat{\boldsymbol{\theta}}$ for the estimator function, without explicitly including the dependence on $\boldsymbol{y}$.

The previous definition for an estimator is extremely general, and we can choose estimators that provide completely useless results (e.g. $T(\boldsymbol{y}) = 0 \ \forall \boldsymbol{y}$). We are interested in picking estimators that give "good" estimations of our parameter according to a given criterion. There is a rich theory that studies the properties of such estimators and several classes of estimators that are used in applications. We will focus on one of the most widely used estimators, the so-called *maximum likelihood estimators* .

A central idea in frequentist estimation is that we define *a probabilistic space for the outcomes of our experiment*. The sample points are the observed data $\boldsymbol{y}$. The sample space Ω is the set of all possible observations that we can obtain. For example, if $\boldsymbol{y}_i \in \{0, 1\}$, Ω will be the set of all binary sequences of m bits. For discrete spaces, we define a PMF that gives us the probability of observing a given event, while for continuous spaces, we define a PDF. It is customary to use the same symbol for both cases, as in most cases whether we are using a PMF or a PDF will be clear from the context. These functions would depend on the value of the parameter $\boldsymbol{\theta}$, meaning that we could use a notation such as $p_{\boldsymbol{\theta}}(\boldsymbol{y})$ for such PMF/PDF's (or we can be more specific and use $f_{\boldsymbol{\theta}}(\boldsymbol{y})$ for PDF's). We will also use the notation $p(\boldsymbol{y}|\boldsymbol{\theta}) = p_{\boldsymbol{\theta}}(\boldsymbol{y})$ for the likelihood of the data $\boldsymbol{y}$ given the parameter $\boldsymbol{\theta}$.

1.3.1 Quantifying the difference between two probability distributions

1.3.2 Maximum likelihood estimators

A very common choice for our estimator T is the *maximum likelihood estimator*, defined as:

$$\hat{\boldsymbol{\theta}}_{ML} = T_{ML}(\boldsymbol{y}) = argmax_{\boldsymbol{\theta}}\, p(\boldsymbol{y}|\boldsymbol{\theta}).$$

To clarify ideas, let us consider the following example: Assume that we have 5 coins, with different head probabilities: $\theta_i = i/4, i = 0...4$ (θ is a scalar). We pick one of these coins (with the same probability), and we toss the coin in 3 independent experiments, obtaining the following result: $\boldsymbol{y} = (1, 0, 0)$.

In order to find $\hat{\theta}_{ML}$, we can define 5 different probability spaces, with the same sample space $\Omega = \{(0,0,0), (0,0,1), (0,1,0), (0,1,1), (1,0,0), (1,0,1), (1,1,0), (1,1,1)\}$ but with different probability distributions $p(\boldsymbol{y}|\boldsymbol{\theta})$ with $\theta = i/4,\ i = 0...4$.

The result of our experiment is $\boldsymbol{y} = (1, 0, 0)$, and for each probability space we will have:

- $p((1,0,0)|\boldsymbol{\theta} = 0) = 0$,
- $p((1,0,0)|\boldsymbol{\theta} = \frac{1}{4}) = \frac{1}{4}(\frac{3}{4})^2 = \frac{9}{64}$,
- $p((1,0,0)|\boldsymbol{\theta} = \frac{1}{2}) = \frac{2}{4}(\frac{2}{4})^2 = \frac{8}{64}$,
- $p((1,0,0)|\boldsymbol{\theta} = \frac{3}{4}) = \frac{3}{4}(\frac{1}{4})^2 = \frac{3}{64}$,
- $p((1,0,0)|\boldsymbol{\theta} = 1) = 0$.

and then: $T_{ML}((1,0,0)) = \hat{\boldsymbol{\theta}}_{ML} = \frac{1}{4}$.

1.3.3 Estimators are random variables

Estimators are (measurable) functions of the sample points $\boldsymbol{y}$ of the sample space Ω, with PMF/PDF $p(\boldsymbol{y}|\boldsymbol{\theta})$, meaning that *estimators are random variables*.

To clarify ideas, assume that in our previous example, the true value of the parameter is $\theta = \frac{1}{2}$. Assume also that we have different experimenters, each of them obtaining an experimental data set $\boldsymbol{y}^{(i)}$. Each of the observed values can be different (for instance we can have $\boldsymbol{y}^{(1)} = (1,0,0)$, $\boldsymbol{y}^{(2)} = (0,0,0)$, etc), and the estimated values of the parameters would also be different ($\hat{\boldsymbol{\theta}}^{(1)}_{ML} = \frac{1}{4}$, $\hat{\boldsymbol{\theta}}^{(1)}_{ML} = 0$, etc). In other words, for a fixed value of the parameter θ, we can define a probability distribution function for the random variable $\boldsymbol{\theta}_{ML}$.

In our example, for $\theta = \frac{1}{2}$, we would have:

- $p_{\hat{\boldsymbol{\theta}}_{ML}}(0) = p(\hat{\boldsymbol{\theta}}_{ML} = 0) = p((0,0,0)) = (\frac{1}{2})^3 = \frac{1}{8}$,
- $p_{\hat{\boldsymbol{\theta}}_{ML}}(\frac{1}{4}) = p(\hat{\boldsymbol{\theta}}_{ML} = \frac{1}{4}) = p(\boldsymbol{y} \in \{(1,0,0),(0,1,0),(0,0,1)\}) = \frac{3}{8}$,
- $p_{\hat{\boldsymbol{\theta}}_{ML}}(\frac{1}{2}) = p(\hat{\boldsymbol{\theta}}_{ML} = \frac{1}{2}) = 0$,
- $p_{\hat{\boldsymbol{\theta}}_{ML}}(\frac{3}{4}) = p(\hat{\boldsymbol{\theta}}_{ML} = \frac{3}{4}) = p(\boldsymbol{y} \in \{(1,1,0),(1,0,1),(0,1,1)\}) = \frac{3}{8}$,
- $p_{\hat{\boldsymbol{\theta}}_{ML}}(1) = p(\hat{\boldsymbol{\theta}}_{ML} = 1) = p(\boldsymbol{y} = (1,1,1)) = (\frac{1}{2})^3 = \frac{1}{8}$,

It is interesting to see that in this example, the ML estimator never produces the correct value of θ.

1.3.4 Maximum likelihood estimation of Bernoulli random variables

We can solve now our example estimation problem for a more general case of a Bernoulli random variable.

We find the general expression for the likelihood function as:

$$p(\boldsymbol{y}|\theta) = \theta^{\sum y_i}(1-\theta)^{m-\sum y_i}.$$

For finding the maximum likelihood estimator, we must find the value of θ that maximizes this function for a given $\boldsymbol{y}$. Very often, we maximize the logarithm of this function, the so-called *log-likelihood*:

$$\log p(\boldsymbol{y}|\theta) = \sum y_i \log(\theta) + (m - \sum y_i)\log(1-\theta).$$

As $log(x)$ is an increasing function, the value of $\theta \in [0,1]$ that maximizes the log-likelihood is the same as the one that maximizes the likelihood function. In this simple case, we can obtain the maximum by derivation:

$$\frac{d}{d\theta}\log p(\boldsymbol{y}|\theta) = \frac{\sum y_i}{\theta} - \frac{m - \sum y_i}{1-\theta} = 0.$$

which leads to $\hat{\theta}_{ML} = \frac{\sum y_i}{m}$.

1.3.5 Expected value and variance of an estimator.

The expected value of the estimator T, $\mathbb{E}(\hat{\boldsymbol{\theta}})$, and its variance, $Var(\hat{\boldsymbol{\theta}})$, are defined across all possible outcomes of our experiment, keeping $\boldsymbol{\theta}$ constant, and in general they will be a function of $\boldsymbol{\theta}$:

$$\mathbb{E}_{\hat{\boldsymbol{\Theta}}}(\hat{\boldsymbol{\theta}}) = \int \hat{\boldsymbol{\theta}}\, p_{\hat{\boldsymbol{\Theta}}}(\hat{\boldsymbol{\theta}}) d\hat{\boldsymbol{\theta}}.$$

In the previous integral, we assume that we know the distribution of the estimator $\hat{\boldsymbol{\theta}}$, which can, in general, be obtained taking into account the distribution of $\boldsymbol{y}$ and the functional form of the estimator according to the relationship $\hat{\boldsymbol{\Theta}} = T(\boldsymbol{Y})$. We can then write:

$$\mathbb{E}_{\hat{\boldsymbol{\Theta}}}(\hat{\boldsymbol{\theta}}) = \mathbb{E}_{\boldsymbol{Y}}(\hat{\boldsymbol{\theta}}(\boldsymbol{y})) = \int T(\boldsymbol{y})\, p_{\boldsymbol{Y}}(\boldsymbol{y}) d\boldsymbol{y}.$$

Very often we use simply the symbol $\mathbb{E}(\hat{\boldsymbol{\theta}})$.

The *bias* of the estimator is defined as $b(\hat{\boldsymbol{\theta}}) = |\,\mathbb{E}(\hat{\boldsymbol{\theta}}) - \boldsymbol{\theta}|$. If for all values of $\boldsymbol{\theta}$ we have $b(\hat{\boldsymbol{\theta}}) = 0$ we say that our estimator is *unbiased*.

In our previous example, the maximum likelihood estimator has the expression:

$$\hat{\theta}_{ML} = \frac{\sum y_i}{m}.$$

As the data y_i follows a Bernoulli distribution, we deduce that the distribution of our estimator will be:

$$p_{\hat{\theta}_{ML}}(\frac{k}{m}) = \binom{m}{k} \theta^k (1-\theta)^{m-k}.$$

Regarding expected value and variance:

$$\mathbb{E}(\hat{\theta}_{ML}) = \frac{\sum \mathbb{E}(y_i)}{m} = \mathbb{E}(y_1) = \theta,$$
$$Var(\hat{\theta}_{ML}) = \frac{\sum Var(y_i)}{m^2} = \frac{Var(y_1)}{m} = \frac{\theta(1-\theta)}{m}.$$

we then see that our maximum likelihood is, in this case, unbiased and that its variances go to zero for large values of m.

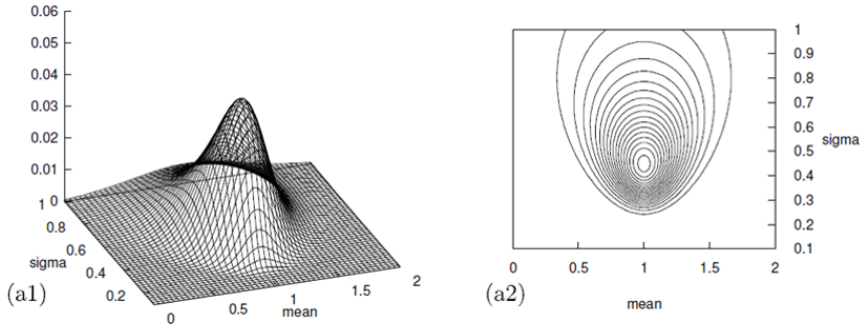
1.3.6 ML estimation of the expected value and variance of a Gaussian random variable

Assume now that Y follows a Gaussian distribution of unknown expected value μ and variance σ^2 (in other words, we estimate the parameter $\boldsymbol{\theta} = (\mu, \sigma)$). We take m independent samples, obtaining a vector $\boldsymbol{y} = (y_1, ..., y_m)$. Our goal, is to estimate the value of $\boldsymbol{\theta} = (\mu, \sigma^2)$ using Maximum Likelihood estimation.

As we have seen previously, the likelihood of our data given the value of the parameter θ is:

$$p(\boldsymbol{y}|\mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{m/2}} \prod_i e^{-\frac{(y_i-\mu)^2}{2\sigma^2}} = \frac{1}{(2\pi\sigma^2)^{m/2}}\, e^{-\frac{\sum_i (y_i-\mu)^2}{2\sigma^2}}.$$

This is a two-variable function that also depends on the values of the results of the experiment y_i:



Often, it is convenient to work with the *log-likelihood* function instead:

$$\log p(\boldsymbol{y}|\mu,\sigma^2) = -\frac{m}{2}\log(2\pi) - \frac{m}{2}\log(\sigma^2) - \frac{\sum_i (y_i-\mu)^2}{2\sigma^2}.$$

The maximum of the likelihood function can be found by simply obtaining the expression of the gradient of the log-likelihood function and finding the values of μ and σ^2 that make it equal to the vector zero:

$$\nabla_{\mu,\sigma^2} \log p(\boldsymbol{y}|\mu,\sigma^2) = (0,0).$$

The two equations for the components of the gradient are:

$$\frac{\partial \log p(\boldsymbol{y}|\mu,\sigma^2)}{\partial \mu} = \frac{\sum_i (y_i-\mu)}{\sigma^2} = \frac{\sum_i y_i}{\sigma^2} - \frac{m\mu}{\sigma^2} = 0,$$

$$\frac{\partial \log p(\boldsymbol{y}|\mu,\sigma^2)}{\partial \sigma^2} = -\frac{m}{2\sigma^2} + \frac{\sum_i (y_i-\mu)^2}{2\sigma^4} = 0.$$

Note that these equations *must be fulfilled simultaneously.*

It is easy to find that the solution is:

- $\hat{\mu} = \mu_{ML} = \frac{\sum_i y_i}{m}$, *independently of the value of* σ^2_{ML}.
- $\hat{\sigma^2} = \sigma^2_{ML} = \frac{\sum_i (y_i - \mu_{ML})^2}{m}$, which obviously depends on μ_{ML}.

Using standard techniques in calculus (e.g. negative definiteness of the Hessian matrix) one finds that this critical point effectively corresponds to a maximum.

Distribution of the estimator for the mean

From the expression of the ML estimator μ_{ML} we easily deduce that its distribution must be Gaussian (since it is the sum of m independent Gaussian random variables Y/m), with $\mathbb{E}_{\boldsymbol{Y}}(\mu_{ML}) = \mu$ and $Var_{\boldsymbol{Y}}(\mu_{ML}) = \sigma^2/m$ (here we stress in the notation that the expected value and variance are taken for different possible observed vectors $\boldsymbol{y}$). In other words, this is good as you can get: μ_{ML} is an *unbiased* estimator with a Gaussian distribution and variance that goes to zero for large values of m (the number of samples).

Distribution of the estimator for the variance

In this case, we can also obtain the exact expression of the distribution of the ML estimator for the variance, which is a χ-*square distribution*. However, what is more interesting for us is to find the expected value of this estimator:

$$\mathbb{E}_{\boldsymbol{Y}}(\sigma^2_{ML}) = \mathbb{E}_{\boldsymbol{Y}}(\frac{\sum_i (y_i - \mu_{ML})^2}{m}) = m\,\mathbb{E}_{\boldsymbol{Y}}(\frac{y_1^2}{m}) - 2m\,\mathbb{E}_{\boldsymbol{Y}}(\frac{y_1 \mu_{ML}}{m}) + m\,\mathbb{E}_{\boldsymbol{Y}}(\frac{\mu^2_{ML}}{m}) =$$

$$= \mathbb{E}_{\boldsymbol{Y}}(y_1^2) - 2\,\mathbb{E}_{\boldsymbol{Y}}(\frac{y_1^2 + y_1 \sum_{i>1} y_i}{m}) + \mu^2 + \frac{\sigma^2}{m} =$$

$$= \sigma^2 + \mu^2 - 2(\frac{\mu^2 + \sigma^2 + \mu^2(m-1)}{m}) + \mu^2 + \frac{\sigma^2}{m} =$$

$$= \sigma^2 + \mu^2 - 2(\frac{\sigma^2}{m} + \mu^2) + \mu^2 + \frac{\sigma^2}{m} = \frac{m-1}{m}\sigma^2.$$

This means that the estimator is *biased* and that on average (if we use many different datasets to obtain many different estimates) we are underestimating its value. We see, however, that for large values of m, the bias $|\sigma^2_{ML} - \sigma^2|$ goes to zero, as it corresponds to an ML estimator.

1.3.7 ML estimation of the expected value and variance of a multivariate Gaussian random variable

For a multivariate Gaussian distribution, the *log-likelihood* function is:

$$\log p(\boldsymbol{y}|\boldsymbol{\mu}, \Sigma) = -\frac{mn}{2}\log(2\pi) - \frac{m}{2}\log|\Sigma| - \frac{1}{2}\sum(\boldsymbol{y}_i - \boldsymbol{\mu})^t \Sigma^{-1}(\boldsymbol{y}_i - \boldsymbol{\mu}).$$

We obtain our ML estimators by solving:

$$\nabla_{\boldsymbol{\mu},\Sigma} \log p(\boldsymbol{y}|\boldsymbol{\mu}, \Sigma) = (0, ..., 0),$$

across all the components of $\boldsymbol{\mu}$ and Σ.

We first find the derivative with respect to the components of $\boldsymbol{\mu}$, expressed as a vector of the components of n:

$$\frac{\partial \log p(\boldsymbol{y}|\boldsymbol{\mu}, \Sigma)}{\partial \boldsymbol{\mu}} = \Sigma^{-1} \sum_i (\boldsymbol{y}_i - \boldsymbol{\mu}) = \Sigma^{-1}(-m\boldsymbol{\mu} + \sum_i \boldsymbol{y}_i) = 0,$$

which leads to:

$$\boldsymbol{\mu}_{ML} = \frac{1}{m} \sum_i \boldsymbol{y}_i,$$

as we assume that Σ^{-1} is positive definite.

For the derivative with respect all the components of Σ, expressed now as an $n \times n$ matrix:

$$\frac{\partial \log p(\boldsymbol{y}|\boldsymbol{\mu}, \Sigma)}{\partial \Sigma} = -\frac{m}{2}\Sigma^{-1} + \frac{1}{2}\Sigma^{-1} \sum_i (\boldsymbol{y}_i - \boldsymbol{\mu})(\boldsymbol{y}_i - \boldsymbol{\mu})^t \Sigma^{-1},$$

where we have used:

$$\frac{\partial \log |\Sigma|}{\partial \Sigma} = (\Sigma^t)^{-1} = \Sigma^{-1},$$

$$\sum_i (\boldsymbol{y}_i - \boldsymbol{\mu})^t \Sigma^{-1} (\boldsymbol{y}_i - \boldsymbol{\mu}) = Tr(\sum_i (\boldsymbol{y}_i - \boldsymbol{\mu})^t \Sigma^{-1} (\boldsymbol{y}_i - \boldsymbol{\mu})) =$$

$$Tr(\Sigma^{-1} \sum_i (\boldsymbol{y}_i - \boldsymbol{\mu})(\boldsymbol{y}_i - \boldsymbol{\mu})^t),$$

and:

$$\frac{\partial Tr(\Sigma^{-1} \sum_i (\boldsymbol{y}_i - \boldsymbol{\mu})(\boldsymbol{y}_i - \boldsymbol{\mu})^t)}{\partial \Sigma} = \Sigma^{-1} \sum_i (\boldsymbol{y}_i - \boldsymbol{\mu})(\boldsymbol{y}_i - \boldsymbol{\mu})^t \Sigma^{-1}.$$

Setting these derivatives to zero, and taking into account that we also need to fulfill the zero in the gradient with respect $\boldsymbol{\mu}$, meaning that we must set $\boldsymbol{\mu} = \boldsymbol{\mu}_{ML}$:

$$\frac{1}{2}(-mI + \Sigma^{-1} \sum_i (\boldsymbol{y}_i - \boldsymbol{\mu}_{ML})(\boldsymbol{y}_i - \boldsymbol{\mu}_{ML})^t)\Sigma^{-1} = 0,$$

we obtain:

$$\Sigma_{ML} = \frac{1}{m} \sum_i (\boldsymbol{y}_i - \boldsymbol{\mu}_{ML})(\boldsymbol{y}_i - \boldsymbol{\mu}_{ML})^t,$$

where we use again the assumption that Σ^{-1} is positive definite, meaning that $A\Sigma^{-1} = 0$ implies $A = 0$.

We can easily see that:

$$\mathbb{E}(\boldsymbol{\mu}_{ML}) = \boldsymbol{\mu},$$

(i.e. it is an unbiased estimator) and

$$\mathbb{E}(\Sigma_{ML}) = \frac{m-1}{m}\Sigma,$$

(i.e. it is a biased estimator).

1.3.8 Optional: Minimum Variance Unbiased Estimators (MVUE) and the Rao-Cramer bound

Assume that $\hat{\theta}$ is an *unbounded* estimator. Under general conditions (called *regularity conditions*), we can establish a bound for the variance of $\hat{\theta}$:

$$Var_{\boldsymbol{Y}}(\hat{\theta}) \geq \frac{1}{I(\theta)},$$

where $I(\theta)$ is called the *Fisher information*, defined as:

$$I(\theta) = \frac{1}{m}\,\mathbb{E}_{\boldsymbol{Y}}[(\frac{\partial log(p(\boldsymbol{y}|\theta))}{\partial\theta})^2] = -\frac{1}{m}\,\mathbb{E}_{\boldsymbol{Y}}[\frac{\partial^2 log(p(\boldsymbol{y}|\theta))}{\partial^2\theta}].$$

This result is known as the *Rao-Cramer bound*.

For instance, if Y follows a Gaussian distribution, the ML estimator μ_{ML} is unbiased (i.e. $\mathbb{E}(\mu_{ML}) = \mu$), and has variance $Var(\mu_{ML}) = \frac{\sigma^2}{m}$. In this case, we fulfill the required regularity conditions, and we have: $I(\mu) = \frac{m}{\theta^2}$.

This means that in this case μ_{ML} is the *Minimum Variance Unbounded Estimator (MVUE)* of μ.

The Rao-Cramer bound can be extended to the multivariate case (i.e. when $\boldsymbol{\theta}$ is a vector). In this case, the Fisher information is the symmetric matrix with components:

$$I_{ij}(\boldsymbol{\theta}) = -\frac{1}{m}\,\mathbb{E}_{\boldsymbol{Y}}[\frac{\partial^2 log(p(\boldsymbol{y}|\theta))}{\partial\boldsymbol{\theta}_i\partial\boldsymbol{\theta}_j}], \tag{1.3.8.1}$$

and the bound is expressed as a positive definite inequality between the variance-covariance matrix of $\hat{\boldsymbol{\theta}}$ and the inverse of the Fisher information matrix.

1.3.9 Second order approximation of the likelihood around $\boldsymbol{\theta}_{ML}$

If $\log p(\boldsymbol{y}|\boldsymbol{\theta})$ is a smooth differentiable function, we can use the following Taylor expansion in a neighborhood of $\boldsymbol{\theta}_{ML}$:

$$\log p(\boldsymbol{y}|\boldsymbol{\theta}) = \log p(\boldsymbol{y}|\boldsymbol{\theta}_{ML}) - \frac{m}{2}(\boldsymbol{\theta}-\boldsymbol{\theta}_{ML})^t I(\boldsymbol{\theta}_{ML})(\boldsymbol{\theta}-\boldsymbol{\theta}_{ML}) + o(|\boldsymbol{\theta}-\boldsymbol{\theta}_{ML}|^2).$$

1.3.10 Properties of Maximum Likelihood estimators

Much of the popularity of maximum likelihood estimators is due to their properties for large sample sizes. Under some conditions (not stated here), we have the following results:

- Although in our previous example, we obtained an unbiased estimator, *ML estimators are in general biased*. However, for large samples, the ML estimator becomes unbiased and has the minimum possible variance (i.e. it becomes the Minimum Variance Unbiased Estimator, MVUE).
- For large samples, the distribution of $\hat{\boldsymbol{\theta}}_{ML}$ becomes closer to a Gaussian of average $\boldsymbol{\theta}$, and a variance that can be computed using the *Fisher's information*.
- ML estimators are *consistent*, meaning that for large samples, $\hat{\boldsymbol{\theta}}_{ML} \to \boldsymbol{\theta}$ in probability. In other words, as the number of observations increases, the distribution of the maximum likelihood estimator becomes more and more concentrated on the value $\boldsymbol{\theta}$.

The number of samples required for achieving these asymptotic properties depends on the dimensionality of the parameter $\boldsymbol{\theta}$. In general, if the number of dimensions of the vector $\boldsymbol{\theta}$ increases, the required value of m can become extremely high, a phenomenon which is a consequence of the so-called *the curse of dimensionality*.

1.4 Optional: KL-divergence, cross-entropy and maximum likelihood

Assume that $p_Y(y) = p(y)$ is the probability distribution of the random variable Y. We want to approximate this distribution by a parametric distribution $p_\theta(y)$.

One way of formulating this problem would be to find the θ that **minimizes the KL-divergence** between the two distributions (pay attention to the order!):

$$\min_\theta D_{KL}(p(y)||p_\theta(y)) = H(Y) + \min_\theta \mathbb{E}_{p(y)}(-\log p_\theta(y)),$$

which is equivalent to **minimizing the cross-entropy** between the actual and the parametric distributions.

Instead of minimizing the expected value we can minimize the *empirical* cross entropy, where $y_i \in \mathcal{D}$:

$$\min_\theta \ \mathbb{E}_{p(y)}(-\log p_\theta(y)) = \min_\theta \ -\frac{1}{T}\sum_{i=1}^{T} \log p_\theta(y_i),$$

which is **equivalent to maximum (log) likelihood estimation**:

$$\max_{\theta} \log \prod_{i}^{T} p_\theta(y_i) = \max_{\theta} \log p_\theta(y_1, ..., y_T).$$

1.4.1 Gaussian distribution

1.4.2 Categorical distributions

The cross-entropy between the empirical distribution for $p(1)$, i.e. $\frac{k}{m}$, and our estimator for $p(1)$, i.e. $\hat{\theta}$, is:

$$H(\frac{\sum y_i}{m}, \hat{\theta}) = \frac{\sum y_i}{m} \log(\hat{\theta}) + (1 - \frac{\sum y_i}{m}) \log(1 - \hat{\theta}).$$

We see then that minimizing the log-likelihood is equivalent to minimizing the cross-entropy between the empirical distribution and our estimated distribution.

1.5 Unbiased estimators. The bias-variance tradeoff. Mean Square Error (MSE)

The bias ($b(\hat{\boldsymbol{\theta}})$) and the variance ($Var(\hat{\boldsymbol{\theta}})$) are two important characterizations of an estimator. For instance, we have seen that we characterize some estimators as the MVUE, the unbiased estimator with minimum variance.

Another important characteristic of an estimator is its *Mean Square Error(MSE)*, i.e. the expected value of the squared error between estimation and the true value of the parameter. If the parameter θ is a scalar, the MSE is defined as:

$$MSE(\hat{\theta}) = \mathbb{E}_{\hat{\Theta}}(|\hat{\theta} - \theta|^2),$$

while for a vector parameter:

$$MSE(\hat{\boldsymbol{\theta}}) = \mathbb{E}_{\hat{\boldsymbol{\Theta}}}((\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})^t(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})).$$

When comparing different estimators, it is very often convenient to determine the one that has a smaller MSE. In these cases it is interesting to consider the following equality:

$$MSE(\hat{\boldsymbol{\theta}}) = b(\hat{\boldsymbol{\theta}})^2 + Var(\hat{\boldsymbol{\theta}}),$$

which expresses what is called the *bias-variance tradeoff*, which says that *sometimes we are interested in using a biased estimator provided that has a very small variance, as it can lead to a smaller MSE.*

We will see that this equality plays also an important role in comparing supervised Machine Learning (ML) methods.

The derivation of the previous equation for a scalar parameter is:

$$MSE(\hat{\theta}) = \mathbb{E}_{\hat{\Theta}}(|\hat{\theta}-\theta|^2) = \mathbb{E}_{\hat{\Theta}}(\hat{\theta}^2-2\hat{\theta}\theta+\theta^2) = \mathbb{E}_{\hat{\Theta}}(\hat{\theta}^2-2\hat{\theta}\theta+\theta^2+\mathbb{E}^2_{\hat{\Theta}}(\hat{\theta})-\mathbb{E}^2_{\hat{\Theta}}(\hat{\theta})) =$$

$$\mathbb{E}_{\hat{\Theta}}(\hat{\theta}^2) - 2\,\mathbb{E}_{\hat{\Theta}}(\hat{\theta})\theta + \theta^2 + \mathbb{E}^2_{\hat{\Theta}}(\hat{\theta}) - \mathbb{E}^2_{\hat{\Theta}}(\hat{\theta}) = Var(\hat{\theta}) + \mathbb{E}^2_{\hat{\Theta}}(\hat{\theta}) + \theta^2 - 2\,\mathbb{E}_{\hat{\Theta}}(\hat{\theta})\theta =$$

$$Var(\hat{\theta}) + b^2(\hat{\theta}).$$

This proof can be easily extended for the case of a vector parameter.

1.6 Linear regression: Linear estimation for a linear model

We can extend our previous formulation of the estimation problem to the case in which we have two joint random variables $\boldsymbol{X}$ (a random vector with components p) and Y characterized by a joint PMF or PDF $p_{\boldsymbol{\theta}}(\boldsymbol{X}, Y)$. We assume that we sample $(\boldsymbol{X}, Y)$ in m independent experiments, obtaining a set of observed data: $\{(\boldsymbol{x}_1, y_1), ..., (\boldsymbol{x}_m, y_m)\}$, that we will call *training set (TS)*.

A very important case appears when we have a linear dependency between the random variables $\boldsymbol{X}$ and Y, in the form:

$$Y = \boldsymbol{X}^t\boldsymbol{\theta} + E,$$

where E is a zero-mean Gaussian random variable, that we will call *error* term or *noise*.

We can further generalize the previous expression assuming a function $\phi(\boldsymbol{x}) = (\phi_1(\boldsymbol{x}), ..., \phi_p(\boldsymbol{x}))$:

$$Y = \phi^t(\boldsymbol{X})\boldsymbol{\theta} + E,$$

where $\phi_i(x)$ are functions of the input parameter $\boldsymbol{x}$. A common choice for ϕ_i is $\phi_i(x) = x^{i-1}$, (i.e. polynomial functions of x, assumed here to be a scalar), but other choices are possible.

The $m \times p$ matrix obtained by considering the m rows $\boldsymbol{\phi}(\boldsymbol{x}_i)^t$ is often called the *design matrix*:

$$\Phi(\boldsymbol{x}) = \begin{bmatrix} \boldsymbol{\phi}^t(\boldsymbol{x}_1) \\ \vdots \\ \boldsymbol{\phi}^t(\boldsymbol{x}_m) \end{bmatrix}.$$

We assume that $m \geq p$ and that the p columns of $\Phi(\boldsymbol{x})$ are linearly independent (i.e. $\Phi(\boldsymbol{x})$ is a full-rank matrix).

The functional relation between parameter $\boldsymbol{\theta}$ and $\boldsymbol{y}$ is linear:

$$\boldsymbol{y} = \Phi(\boldsymbol{x})\boldsymbol{\theta} + \boldsymbol{\epsilon},$$

where $\boldsymbol{\epsilon}$ is an *error vector* of m components. We say thus that we are using a **linear model**.

The problem of **linear regression** consists of giving a *linear estimation* $\hat{\boldsymbol{\theta}}$ for our linear model, given a training set.

Linear estimation implies that the estimation function is a linear function of the input data, i.e. $\hat{\boldsymbol{\theta}} = A\boldsymbol{y}$.

Note: we can use a non-linear estimator to estimate the parameter for a linear model, and we can use a linear estimator to estimate the parameter of a linear model. In other words: *Linear model* $\neq$ *linear estimator*.

1.6.1 Ordinary least squares linear regression (purely geometrical approach)

We study a first approach for the linear regression problem, in which we do not use the probabilistic model formulated before. Instead, we will use some basic results from linear algebra to find the estimation of $\boldsymbol{\theta}$.

The linear regression problem is formulated as follows: *Find the vector* $\boldsymbol{\theta}$ *that minimizes the euclidean norm of the error vector.* The euclidean norm of $\boldsymbol{\epsilon}$ is given by :

$$||\boldsymbol{\epsilon}||_2 = \sqrt{\epsilon_1^2 + ... + \epsilon_n^2},$$

and this optimization criterion is often known as minimizing the *least squares error*.

Alternatively, we may write this condition as finding $\boldsymbol{\theta}$ that minimizes the expression:

$$\min_{\boldsymbol{\theta}} ||\boldsymbol{y} - \Phi(\boldsymbol{x})\boldsymbol{\theta}||_2 = \sum_{i=1}^{m} (y_i - \boldsymbol{\theta}^t \boldsymbol{\phi}(\boldsymbol{x}_i))^2.$$

We have a non-zero error vector $\boldsymbol{\epsilon}$, meaning that the most likely situation is that there is no $\boldsymbol{\theta}$ for which the error can be made zero, which is equivalent to saying that the vector $\boldsymbol{y}$ does not fall into $Col(\Phi(\boldsymbol{x}))$.

There is a well-known result in linear algebra, known as the *projection theorem* which says that the vector belonging to the linear subspace generated by the columns of the matrix $\Phi(\boldsymbol{x})$ which is closer to another arbitrary vector $\boldsymbol{y}$, according to the Euclidean distance, is the orthogonal projection of the vector $\boldsymbol{y}$ into the subspace.

This is equivalent to saying that to minimize error length, *the error vector must be orthogonal to* $Col(\Phi(\boldsymbol{x}))$, i.e. it must belong to $Nul(\Phi(\boldsymbol{x})^t)$. This condition can be expressed as:

$$\Phi(\boldsymbol{x})^t \boldsymbol{\epsilon} = \mathbf{0}.$$

Using the previous condition we arrive to:

$$\Phi(\boldsymbol{x})^t \boldsymbol{\epsilon} = \Phi(\boldsymbol{x})^t (\boldsymbol{y} - \Phi(\boldsymbol{x})\boldsymbol{\theta}) = \mathbf{0},$$

and from here we obtain the expression for the vector $\hat{\boldsymbol{\theta}}_{OLS}$:

$$\hat{\boldsymbol{\theta}}_{OLS} = [\Phi(\boldsymbol{x})^t \Phi(\boldsymbol{x})]^{-1} \Phi(\boldsymbol{x})^t \boldsymbol{y}.$$

The expression $[\Phi(\boldsymbol{x})^t \Phi(\boldsymbol{x})]^{-1} \Phi(\boldsymbol{x})^t$ corresponds to the pseudo-inverse $\Phi(\boldsymbol{x})^+$ when $n \geq p$ and $\Phi(\boldsymbol{x})$ is full rank.

1.6.2 Maximum Likelihood linear regression

In the second approach, we consider the following relationship among the random variables X and Y:

$$Y = \phi^t(\boldsymbol{X})\boldsymbol{\theta} + E = \phi_1(\boldsymbol{X})\boldsymbol{\theta}_1 + ... + \phi_p(\boldsymbol{X})\theta_p + E,$$

and E is a random noise or error that follows a Gaussian distribution $N(0, \sigma^2)$.

Our goal is to give the ML estimators for $\boldsymbol{\theta}$ and σ^2 given the training set TS: i.e. $\hat{\boldsymbol{\theta}}_{ML}$, $\hat{\sigma^2}_{ML}$.

It is important to recall *that TS is a random variable*, meaning that the estimators $\hat{\boldsymbol{\theta}}_{ML}$, $\hat{\sigma^2}_{ML}$ are also random variables.

The likelihood function that we want to optimize is:

$$p(\boldsymbol{y}|\boldsymbol{x}, \boldsymbol{\theta}, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{m/2}} e^{-\frac{\sum_i (y_i - \boldsymbol{\theta}^t \boldsymbol{\phi}(\boldsymbol{x}_i))^2}{2\sigma^2}},$$

or equivalently, the log-likelihood function:

$$\log p(\boldsymbol{y}|\boldsymbol{x}, \boldsymbol{\theta}, \sigma^2) = -\frac{m}{2}\log 2\pi - \frac{m}{2}\log \sigma^2 - \frac{\sum_i (y_i - \boldsymbol{\theta}^t \boldsymbol{\phi}(\boldsymbol{x}_i))^2}{2\sigma^2}.$$

Let us take a closer look at the sum term. We found a similar expression when we studied the ML estimation of a Gaussian variable, with the difference that in this case the parameter to estimate (μ) was the same in all the terms in the sum in the exponential, while here for each term of the sum we have a different expected value: $\boldsymbol{\theta}^t \boldsymbol{\phi}(\boldsymbol{x}_i)$. The optimal value of $\boldsymbol{\theta}$ will be independent of the value σ^2, and will be the one that minimizes the term:

$$\sum_{i=1}^{m} (y_i - \boldsymbol{\theta}^t \boldsymbol{\phi}(x_i))^2.$$

We recognize that this term also appeared in the geometric least square linear regression, meaning that:

$$\hat{\boldsymbol{\theta}}_{ML} = \hat{\boldsymbol{\theta}}_{OLS} = [\Phi(\boldsymbol{x})^t \Phi(\boldsymbol{x})]^{-1} \Phi(\boldsymbol{x})^t \boldsymbol{y},$$

Finally, once we know $\hat{\boldsymbol{\theta}}_{ML}$, we obtain:

$$\hat{\sigma^2}_{ML} = \frac{\sum_i (y_i - \boldsymbol{\theta}_{ML}^t \boldsymbol{\phi}(x_i))^2}{m}.$$

Typically, we will use instead an unbiased estimator for σ^2 given by:

$$\hat{\sigma^2}_{unbiased} = \frac{\sum_i (y_i - \boldsymbol{\theta}_{ML}^t \boldsymbol{\phi}(x_i))^2}{m - p - 1}.$$

1.6.3 Optional: Determination coefficient

Very often we use the determination coefficient R^2 to estimate how good is our linear fit to the data. A related measure is the Fraction of Variance Unexplained, (FVU): $R^2 = 1 - FVU$.

The FVU is defined as the quotient between the (empirical) variance or the MSE of the result of our linear model and variance or the MSE of the best estimation when we do not have any model to fit y (for instance, as we do not know the $\boldsymbol{x}$).

Assume that we do not have any model to fit y. We want to find a estimator $\hat{z}$ of y that minimizes $\hat{MSE}(\hat{z}) : \frac{1}{m} \sum_i (\hat{z} - y_i)^2$. We have:

$$\frac{d}{d\hat{z}} \hat{MSE}(\hat{z}) = \frac{2}{m}(m\hat{z} - \sum_i y_i) = 0,$$

which leads to $\hat{z} = \frac{1}{m} \sum y_i = \hat{E}(y)$ (the empirical mean of y), and $\hat{MSE}(\hat{z}) = \frac{1}{m} \sum (y_i - \hat{E}(y))^2 = \frac{SSTOT}{m}$.

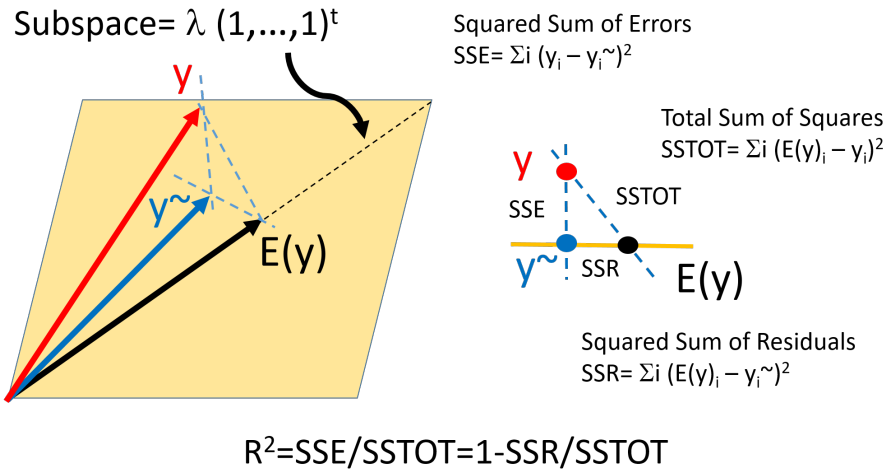
On the other hand, if we use our linear model, we obtain an estimator $\hat{y} = \Phi(\boldsymbol{x})\hat{\theta}$, and $\hat{MSE}(\hat{y}) = \frac{1}{m} \sum (\hat{y}_i - \hat{E}(y))^2 = \frac{SSR}{m}$.

The fraction of variance unexplained $FVU = \frac{SSR}{SSTOT}$ is then the fraction of the empirical MSEs (or equivalently, the fraction of variances) of our linear model and the estimation without any model:

$$R^2 = \frac{SSE}{SSTOT} = \frac{SSTOT - SSR}{SSTOT} = 1 - \frac{SSR}{SSTOT}.$$

The term SSE is proportional to the variance of the errors in the model. SSR is proportional to the variance of errors in our estimation. SSTOT is proportional to the variance of the data (i.e. due to the variability intrinsic to the data).

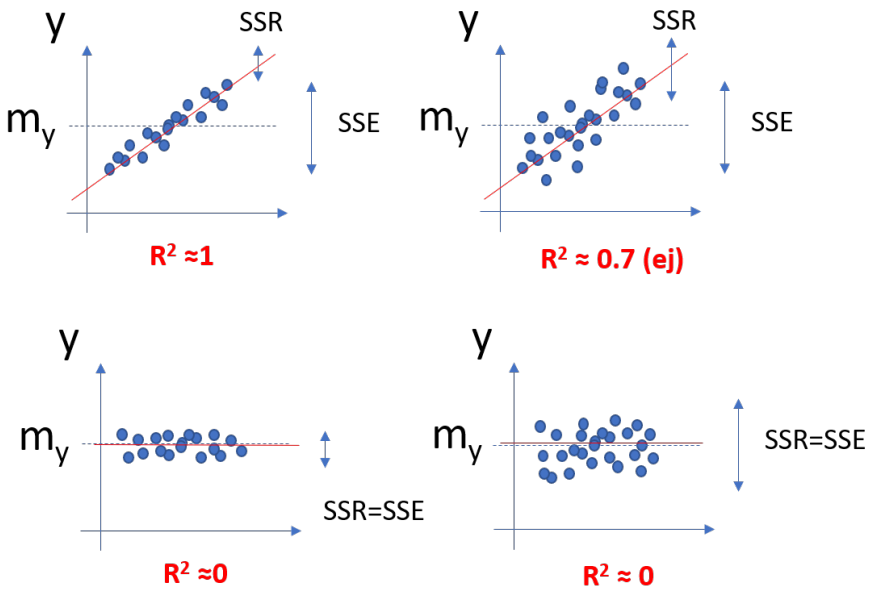
The geometry is shown in the following figure:



$$R^2=SSE/SSTOT=1-SSR/SSTOT$$

A value of $R^2 \approx 1$ means a very good fit of the model (i.e. $FUV \approx 0$). A value $R^2 \approx 0$ means a poor fit of the model (i.e. $FUV \approx 1$).

The following example shows different examples regarding R^2.:



The number of parameters in our model influences the values of R^2, and usually, models with more parameters give better values of R^2. We can take this effect into account if we use unbiased estimators for the variance: $\hat{MSE}(\hat{z}) = \frac{SSTOT}{m-1}$, and $\hat{MSE}(\hat{y}) = \frac{SSR}{m-p}$, obtaining the *Adjusted Coefficient of Determination*,

$$R_a^2 = \frac{SSE(m-1)}{SSTOT(m-p)} = R^2 - \frac{p-1}{m-p}(1-R^2).$$

Alternative ways of assessing the goodness of our estimation are the use of *Information criteria*, such as the *Akaike Information Criterion, AIC* or the *Bayesian Information Criterion, BIC*, that we will see later.

1.6.4 Optional: Best Linear Unbiased Estimator (BLUE): Gauss-Markov theorem

Recall that we are studying *linear* estimators of a linear model Y given $\boldsymbol{X}$ of the form:

$$Y = \phi(\boldsymbol{X})^t\boldsymbol{\theta} + E,$$

where E is the *error* term.

Note: Linear model means that Y isn a linear function of the parameter $\boldsymbol{\theta}$, i.e. $Y = \phi(\boldsymbol{X})^t\boldsymbol{\theta} + E$. But when we say *linear estimators*, we are referring to fact that the estimator function $\hat{\boldsymbol{\theta}} = T(\boldsymbol{y})$ is linear:

$$\hat{\boldsymbol{\theta}} = T(\boldsymbol{y}) = C\boldsymbol{y}.$$

The **Gauss-Markov theorem** establishes that under quite general conditions, **the estimators $\hat{\boldsymbol{\theta}}_{MMSE} = \hat{\boldsymbol{\theta}}_{ML} = \hat{\boldsymbol{\theta}}_{OLS} = [\Phi(\boldsymbol{x})^t\Phi(\boldsymbol{x})]^{-1}\Phi(\boldsymbol{x})^t\boldsymbol{y}$ are the unbiased linear estimators with lower variance, i.e. the Best Linear Unbiased Estimators, (BLUE)**.

Note that we have m samples of $(\boldsymbol{X}, Y)$, and for each of them we have:

$$Y_i = \phi^t(\boldsymbol{X_i})\boldsymbol{\theta} + E_i,\ i = 1, ..., m,$$

which can be established also in vector form using the matrix $\Phi(\boldsymbol{X})$ and error vector $\boldsymbol{E}$.

Some of the conditions under which the G-M theorem holds are the following:

- $\mathbb{E}(E_i) = 0$, i.e. zero-mean errors.
- $Var(E_i) = \sigma^2 < \infty$, i.e. finite and equal variances, a condition called *homoscedasticity*.

- $Var(E_i E_j) = 0$, i.e. uncorrelated errors.

Note that we do not need to have Gaussian errors.

The proof of the G-M theorem is as follows. Assume that we have another unbiased linear estimator

$$\bar{\boldsymbol{\theta}} = C\boldsymbol{y} = ([\Phi(\boldsymbol{x})^t\Phi(\boldsymbol{x})]^{-1}\Phi(\boldsymbol{x})^t+\Delta)\boldsymbol{y} = ([\Phi(\boldsymbol{x})^t\Phi(\boldsymbol{x})]^{-1}\Phi(\boldsymbol{x})^t+\Delta)(\Phi(\boldsymbol{x})\boldsymbol{\theta}+\boldsymbol{e}),$$

where Δ is a matrix. As $\bar{\boldsymbol{\theta}}$ is an unbiased estimator, we have:

$$\mathbb{E}(\bar{\boldsymbol{\theta}}) = \mathbb{E}[[\Phi(\boldsymbol{x})^t\Phi(\boldsymbol{x})]^{-1}\Phi(\boldsymbol{x})^t + \Delta)(\Phi(\boldsymbol{x})\boldsymbol{\theta} + \boldsymbol{e})] =$$

$$\mathbb{E}[[\Phi(\boldsymbol{x})^t\Phi(\boldsymbol{x})]^{-1}\Phi(\boldsymbol{x})^t + \Delta)\Phi(\boldsymbol{x})\boldsymbol{\theta}] = \boldsymbol{\theta} + \Delta\Phi(\boldsymbol{x})\boldsymbol{\theta},$$

which implies $\Delta\Phi(\boldsymbol{x}) = 0$.

For the variance we have:

$$Var(\bar{\boldsymbol{\theta}}) = CVar(\boldsymbol{y})C^t = \sigma^2 CC^t = \sigma^2[\Phi(\boldsymbol{x})^t\Phi(\boldsymbol{x})]^{-1}+\sigma^2\Delta\Delta^t = Var(\hat{\boldsymbol{\theta}})+\sigma^2\Delta\Delta^t.$$

As $\sigma^2\Delta\Delta^t$ is a positive definite matrix, we obtain the PD inequality:

$$Var(\bar{\boldsymbol{\theta}}) \geq Var(\hat{\boldsymbol{\theta}}),$$

meaning that our OLS estimator is the BLUE.

1.7 Optional: Generalizations of Least Squares

1.7.1 Optional: Generalized Least Squares (GLS)

When we have non-homogeneous errors (although we still have finite error variances), we can still find an unbiased linear estimator:

$$\hat{\boldsymbol{\theta}}_{GLS} = [\Phi(\boldsymbol{x})^t\Sigma_\epsilon^{-1}\Phi(\boldsymbol{x})]^{-1}\Phi(\boldsymbol{x})^t\Sigma_\epsilon^{-1}\boldsymbol{y},$$

where Σ_ϵ is the variance-covariance matrix of the error vector $\boldsymbol{E}$. This is known as the *Generalized Least Squares, (GLS)* estimator and it can be shown to be the BLUE for quite general conditions.

1.7.2 Optional: Total Least Squares (TLS)

Total Least Squares are applied when both the variables X and Y are subjected to noise. We assume that these variables are normalized so that $m_X = 0$ and $m_Y = 0$.

Let us study the case of a single parameter X, meaning that we want to find a line l which has the best fit to the set $\{(x_1, y_1), ..., (x_m, y_m)\}$ in the sense that we minimize the quadratic distance from the points (x_i, y_i) to the line l, i.e., solve:

$$min_l \sum_{i=1}^{m} \text{dist}((x_i, y_i), l)^2.$$

We express the equation of the line l as:

$$l = \{(x, y) \in \boldsymbol{R}^2 \mid (a, -1) \begin{pmatrix} x \\ y \end{pmatrix} = 0\}.$$

If (f_i, g_i) is the point in the line l which is at shortest distance of (x_i, y_i) we can also express the distance as:

$$\text{dist}((x_i, y_i), l)^2 = (x_i - f_i)^2 + (y_i - g_i)^2; \;\; af_i - g_i = 0,$$

which leads to the minimization problem:

$$min_{a, \boldsymbol{f}, \boldsymbol{g}} ||\boldsymbol{x} - \boldsymbol{f}||^2 + ||\boldsymbol{y} - \boldsymbol{g}||^2 \quad ,$$

s.t.

$$a\boldsymbol{f} - \boldsymbol{g} = \boldsymbol{0},$$

If we define the matrices $B = \begin{bmatrix} \boldsymbol{x}^t \\ \boldsymbol{y}^t \end{bmatrix}$, and $E = \begin{bmatrix} \boldsymbol{f}^\top \\ \boldsymbol{g}^\top \end{bmatrix}$, we obtain the equivalent optimisation problem:

$$min_{\boldsymbol{f}, \boldsymbol{g}} ||B - E||_F^2 \quad ,$$

s.t.

$$\text{rank}(E) = 1, \;\; (\text{as } (a, -1)E = 0).$$

From the Eckart-Young theorem, the solution is given by the highest singular values and vectors of B:

$$E = \begin{bmatrix} \boldsymbol{f}^\top \\ \boldsymbol{g}^\top \end{bmatrix} = \sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^\top,$$

with $\boldsymbol{u}_1 = \begin{bmatrix} u_{11} \\ u_{12} \end{bmatrix}$, and

$$a = u_{12}/u_{11}.$$

1.7.3 Optional: Nonlinear Least Squares

Linear Least Squares can be expressed as the problem of minimizing

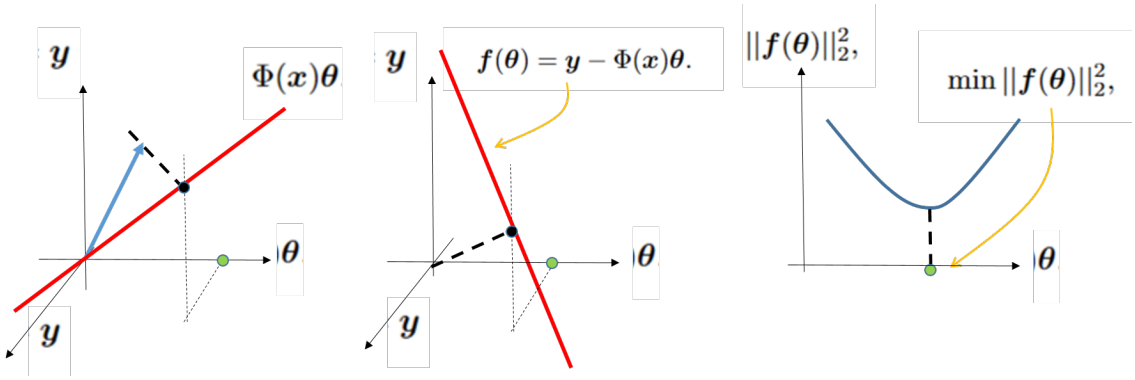
$$\min ||\boldsymbol{f}(\boldsymbol{\theta})||_2^2,$$

where the function $\boldsymbol{f} : \mathbf{R}^p \to \mathbf{R}^m$ is a linear function defined as:

$$\boldsymbol{f}(\boldsymbol{\theta}) = \boldsymbol{y} - \Phi(\boldsymbol{x})\boldsymbol{\theta}.$$

If the matrix $\Phi(\boldsymbol{x})$ has full column rank, the solution is:

$$\boldsymbol{\theta}_{OLS} = \Phi(\boldsymbol{x})^{+}\boldsymbol{y}.$$



Nonlinear least squares generalize this problem for the case in which f is a non-linear differentiable function.

1.7.4 Gauss-Newton method

We define the **Jacobian** of $\boldsymbol{f}$, $J_{\boldsymbol{f}}$ as the matrix function:

$$J_f = \begin{pmatrix} \nabla \boldsymbol{f}_1^t \\ \dots \\ \nabla \boldsymbol{f}_m^t \end{pmatrix}.$$

We linearize $\boldsymbol{f}$ around a value $\boldsymbol{\theta}^*$ as

$$\boldsymbol{f}(\boldsymbol{\theta}) \approx \boldsymbol{f}(\boldsymbol{\theta}^*) + J_{\boldsymbol{f}}(\boldsymbol{\theta} - \boldsymbol{\theta}^*).$$

Moreover, the gradient of $g(\boldsymbol{\theta}) = \min ||\boldsymbol{f}(\boldsymbol{\theta})||_2^2$ is:

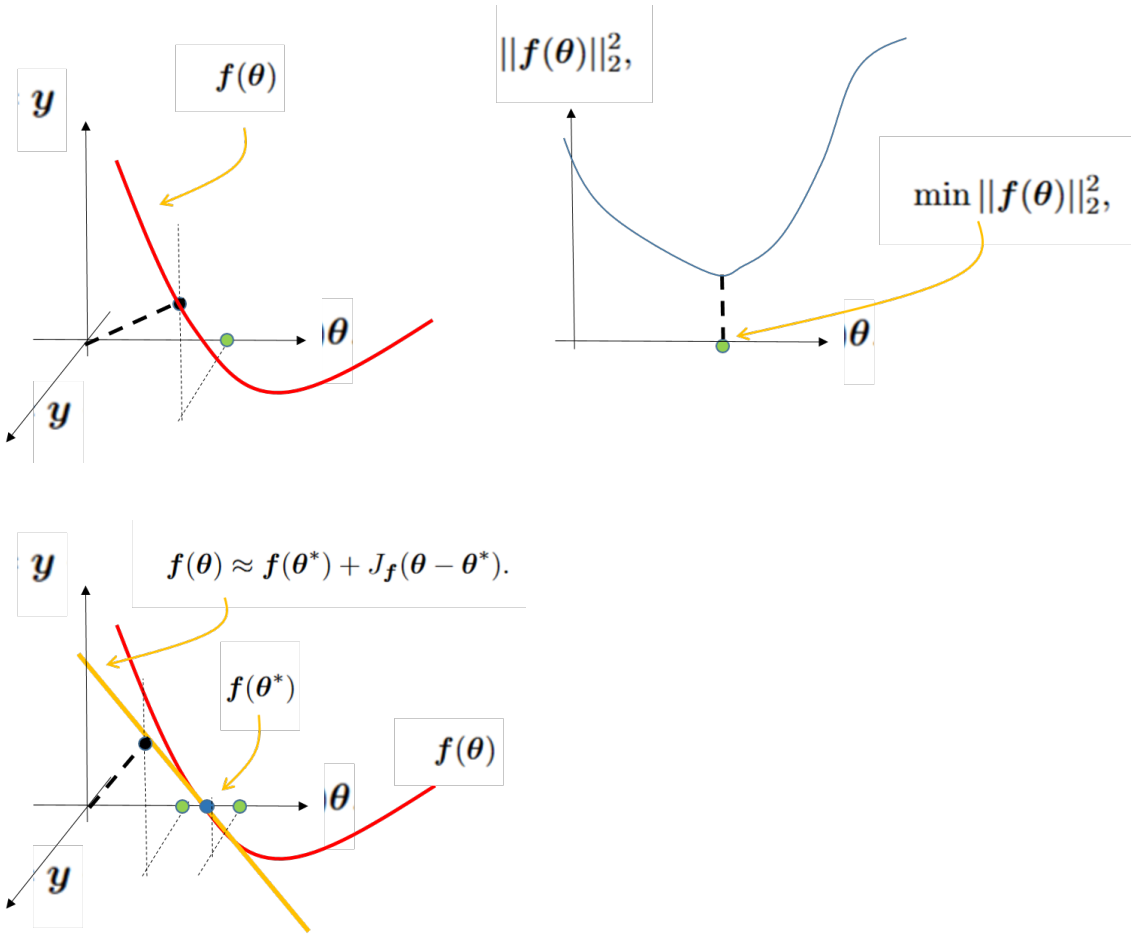
$$\nabla g(\boldsymbol{\theta}) = 2J_{\boldsymbol{f}}^t(\boldsymbol{\theta})\boldsymbol{f}(\boldsymbol{\theta}).$$

The **Gauss-Newton** methods proceed as follows:

1. Start with an initial value $\boldsymbol{\theta}_0$, $k = 0$.
2. Linearize $\boldsymbol{f}$ around $\boldsymbol{\theta}_k$: $\boldsymbol{f}(\boldsymbol{\theta}) \approx \boldsymbol{f}(\boldsymbol{\theta}_k) + J_{\boldsymbol{f}}(\boldsymbol{\theta}_k)(\boldsymbol{\theta} - \boldsymbol{\theta}_k)$.
3. Find the solution to the linear LS problem: $||\boldsymbol{f}(\boldsymbol{\theta}_k) + J_{\boldsymbol{f}}(\boldsymbol{\theta}_k)(\boldsymbol{\theta} - \boldsymbol{\theta}_k)||_2^2$. If $J_{\boldsymbol{f}}(\boldsymbol{\theta}_k)$ is full rank, this is given by:

   $$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - J_{\boldsymbol{f}}(\boldsymbol{\theta}_k)^+ \boldsymbol{f}(\boldsymbol{\theta}_k).$$

   In other words, $\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \boldsymbol{\delta}_k$, where $\boldsymbol{\delta}_k = -J_{\boldsymbol{f}}(\boldsymbol{\theta}_k)^+ \boldsymbol{f}(\boldsymbol{\theta}_k)$.
4. $k = k + 1$, go to step (2), until ∇g is sufficiently small.



1.7.5 Levenberg-Marquardt method

In Gauss-Newton, sometimes the Jacobian is not full column rank. Sometimes the value of $g(\boldsymbol{\theta})$ does not decrease in an iteration. The Levenberg-Marquardt

method uses a regularization of the form:

$$||\boldsymbol{f}(\boldsymbol{\theta}_k) - J_{\boldsymbol{f}}(\boldsymbol{\theta}_k)\boldsymbol{\delta}_k)||_2^2 + \lambda_k||\boldsymbol{\delta}_k||_2^2,$$

which has a solution:

$$\boldsymbol{\theta}^* = (J_{\boldsymbol{f}}(\boldsymbol{\theta}_k)^t J_{\boldsymbol{f}}(\boldsymbol{\theta}_k) + \lambda_k I)^{-1} J_{\boldsymbol{f}}(\boldsymbol{\theta}_k)^t \boldsymbol{f}(\boldsymbol{\theta}_k).$$

λ_k is a regularization factor that is modified in each iteration. The iteration proceeds as follows:

- If $g(\boldsymbol{\theta}^*) < g(\boldsymbol{\theta}_k)$, then $\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}^*$, and $\lambda_{k+1} = \lambda_k \beta_1$, with $\beta_1 < 1$.
- If $g(\boldsymbol{\theta}^*) \geq g(\boldsymbol{\theta}_k)$, then $\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k$, and $\lambda_{k+1} = \lambda_k \beta_2$, with $\beta_2 > 1$,

until ∇g is sufficiently small.

